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STUDY ON THE DEVELOPMENT OF A SOUTH AMERICAN SPACE AGENCY

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Abstract

The idea of establishing a South American Space Agency (SASA) is not new. There have been many discussions about this topic for a while, including an agreement by the South American Nations Union for its acronyms in Spanish UNASUR, to create such space agency. Roughly 10 years ago, Argentina was the first to propose this collaboration with a military orientation. As the ideas progressed, Brazil was proposed as an option to host its headquarters. However, not much support from the South American region has been given in both financial and logistical aspects. To this day such space agency or a similar concept has not yet been established in the region. The Space Generation Advisory Council (SGAC) hosted the first South American Space Generation Workshop (SA-SGW) in Argentina in 2015, where one of the working groups was tasked to further investigate its feasibility, advantages and challenges, in implementing SASA. This paper presents an extension of the main findings from this working group where South American students and young professionals study and present a rationale in favor of SASA, outlining possible solutions and structure that could be taken into account in its implementation. Throughout the paper particular attention is paid to the question: Is it possible for countries in South America to establish such kind of cooperation to stimulate the development and application of capabilities in the space sector, which will allow undertaking missions far beyond the scope of what any single country in South America could do on its own? The existence of a South American Space Agency would allow access to a common orbital vector, which would lower costs, be accessible to all participating countries, and allow engaging with other emerging and established space agencies around the world.

Keywords: South America, Space Agency, SGAC, cooperation, open database.

Acronyms/Abbreviations

ABAE: Agencia Bolivariana de Actividades Espaciales

ACS: Alcantara Cyclone Space

AEP: Agencia Espacial del Paraguay

AOCS: Attitude and Orbit Control Subsystem

CATE: Congreso Argentino de Tecnología Espacial

CBERS: China-Brazil Earth Resources Satellite

CCE: Comisión Colombiana del Espacio

CNOIS: Centro Nacional de Operaciones de Imágenes Satelitales

CONAE: Comisión Nacional de Actividades Espaciales

CONIDA: Comisión Nacional de Investigación y Desarrollo Aeroespacial

EADS: European Aeronautic Defence and Space

ESA: European Space Agency

EXA: Agencia Espacial Civil Ecuatoriana

FAE: Fuerzas Armadas Ecuatorianas

FASat: Fuerza Aérea Satelital de Chile

IGP: Instituto Geofísico del Perú

INPE: Instituto Nacional de Pesquisas Espaciais

ISS: International Space Station

ITA: Instituto Tecnológico da Aeronáutica

LEO: Low Earth Orbit

MECB: Missão Espacial Completa Brasileira

MTCR: Missile Technology Control Regime

NASA: National Aeronautics and Space Agency

PMM: Plataforma Multimissão

SASA: South American Space Agency

SA: South American

SABIAMAR: Satélite Argentino Brasileño para Información del Mar

SA-SGW: South American Space Generation Workshop

SAC: Satélite de Aplicaciones Científicas

SAOCOM: Satellites for Observation and Communications

SGW: Space Generation Workshop

SME: Subject Matter Expert

UNASUR: Unión de Naciones Suramericanas
UN: United Nations
UNCOPUOS: United Nations Committee on the
Peaceful Uses of Outer Space
USA: United States of America
VLS: Vehículo Lanzador de Satélites

1. Introduction

Since the cold war, access to space has been gaining more and more importance, having ceased to serve only as a mean of geopolitical intimidation in the relations among nations, becoming mainly an effective tool in achieving strategic data to governments and the general population. In spite of many applications related to military and defense activities as border control, radar missions, troop displacements, weapon control, etc., most of the countries interested in space activities have been investing their money in civil missions, focused on obtaining public data as urban and crops mapping, telecommunication, earth observation, data collecting system, and positioning systems, within others.

The challenge is that in spite of the price of space technologies getting cheaper, it is still very expensive and even unaffordable for the majority of developing countries in the world. These are found to have access to space services by acquiring space data, services or even entire satellites and ground-segment systems from the developed countries, so they can control their own satellite missions. In South America (SA), this is the case of countries like Venezuela, Bolivia, Chile and Peru.

There are also other countries in South America where space activities are more developed as in Argentina and Brazil. These countries also acquire space services, equipment and entire satellites abroad, but they have their own development, especially in terms of ground-segment and satellites. Currently, both countries also have complete integration and testing facilities in order to develop their own satellites. Regarding launch vehicles, these countries have already developed their own sounding rockets, but they are still incapable to put an object into Earth orbit on their own.

Taking into account these highly heterogeneous characteristics in terms of the development of space technologies in South American countries, any purpose of creating a South American Space Agency (SASA), should contemplate all the countries in the region involved. This means an organization that manages an entire space mission, including ground, space, and transport segments, as well as launch vehicles.

The purpose of this paper is to provide a broad view of the considerations needed for the development of SASA. It looks at the benefits, challenges, possible starting projects the Agency would work on, and some recommendations to address certain challenges

including political, legal, financial, and cultural aspects proper to the South American region.

2. South American Space Generation Regional Workshops

The South American Space Generation Workshops (SA-SGW) initiated in 2015 as part of the Space Generation Advisory Council (SGAC) in support of the United Nations Program on Space Applications. SGAC has been operating for 15 years, and hosting an international Space Generation Workshop (SGW) where students and young professionals from all around the world get together to share ideas and projects about space programs. The regional version of the SGW in South America is held for two consecutive days that includes presentations by subject matter experts (SMEs), group discussions, panels, and working groups covering selected current space topics. During the working groups participants collaborate with peers as well as SMEs and invited guests. The results from the workshops are presented to UNCOPUOS [1]. The first SA-SGW was run in conjunction with the 8th Argentine Congress of Space Technology (CATE for its acronym in Spanish) held in Buenos Aires, Argentina on May 6th-8th, 2015. The working groups discussed the following topics:

1. Education and space outreach programs for South America – how can South America get more involved and stay motivated?
2. Creation of a South American space agency – can/should it be realized?
3. Technology and research in South America and how to support regional advancement – what will it take to grow the South American region?
4. Potential for Mars missions simulations in South America – how/what can South America contribute?

The second SA-SGW took place in Lima, Peru on August 1st-2nd, 2016 as part of the First Peruvian Space Week where the program also included the Latin American Astrobiology Congress and the International GeoBiology Forum. The working groups discussed the following topics:

1. Astrobiology Studies in South America - what is the current status and how can worldwide presence be improved?
2. South American Space Research - Is a common SA Open Space Research Database needed, feasible and realizable?
3. Emerging spacefaring nations – What are viable pathways for their space development?
4. Nanosatellites and Cubesats as an educational and research resource in South America - What venues are currently available to utilize them? What parties

need to be involved and how can they foster the activity?

During both SA-SGW, delegates had a unique opportunity to engage with high-level space sector leaders, professionals and academics. This paper concentrates on the results of the second workshop discussion of the first SA-SGW: “Creation of a South American space agency – can/should it be realized?”. These findings were expanded after the first SGW by some of the members of that working group and other collaborators. As a possible responsibility of this agency, the second topic of the second SA-SGW was proposed: “South American Space Research - is a common SA Open Space Research Database needed, feasible, and realizable?”. Hence, we also cover some of the significant findings of that working group in this paper as well. Descriptions of both topics are shown here for reference [2][3].

2.1 Description of “South American Space Agency – Can/should it be realized?” Working Group (SGAC 1st SA-SGW)

The idea of establishing a South American Space Agency is not new. There have been many discussions about this for a while, and there even was an agreement by the South American Nations Union (UNASUR for its acronyms in Spanish) to create such space agency. The initiative was carried out by an Argentinean general, with a more military oriented collaboration. Brazil was proposed as an option to be the headquarters for this space agency. However, there seemed not to be that much support from the SA region both financially and logistically. Therefore, such space agency has not been established as of today. The existence of a South American Space Agency would allow access to a common orbital vector, which would lower costs and be always accessible. The best method to share costs in a joint effort like this would be to calculate them in relationship to the gross domestic product of each SA nation. The European Space Agency (ESA) has been a successful model for regional space collaboration with its development of Europe’s space capability and the delivery of continuous benefits to the citizens of Europe. Is it possible for countries in South America to establish such kind of cooperation to stimulate the development and application of capabilities in the space sector, which will allow undertaking missions far beyond the scope of what any single country in South America could do on its own?

The Working Group (see Fig. 1) explored challenges experienced on initially creating a South American Space Agency, possibility of establishing a South American Space Agency by 2030s, challenges to organize such Agency, pros/cons to realize such cooperation. The Working Group also proposed recommendations on how to ensure the political and

legal practicalities in the creation of this regional cooperation, best practices for a regional cooperation in the future in case the establishment of such cooperation is concluded not feasible, methods to further cooperation in the space sector among SA countries in the near-term, with or without such Agency.



Fig. 1. First South American Space Generation Workshop – SASA Working Group

2.2 Description of South American Space Research - Is a common SA Open Space Research Database needed, feasible and realizable? (SGAC 2nd SA-SGW)

The spread of Internet allows a large community of researchers and students to be connected. For them, having easy access to state of the art research has a high relevance to develop new technologies and to improve current techniques in different space related fields. The working group should consider whether creating a common database for South American Space Research is needed, exploring existing options around the world. The group should also explore whether the Database is feasible to be created, what entity would manage it, under what legal framework, and how it could sustain its operations monetarily. Last but not least, explain how the database is expected to grow and what venues can be explored to extend its outreach to gather the maximum amount of work possible, creating a true international network.

The Working Group (see Fig. 2) explored who the potential users of the Database would be, who could manage the Database, legal considerations that must be taken to publish papers. The Working Group also proposed recommendations on potential groups interested in pursuing the idea, financial possibilities, and outreach options.



Fig. 2. Second South American Space Generation Workshop – Open Research Database Working Group

3. Historical Background & Current State of Affairs

Space activities in the region go back several decades, with Argentina and Brazil as regional pioneers of the development of sounding rockets by each country's air forces, and first national organizations devoted to space related activities. These two nations have had time enduring space programs, particularly in the case of Brazil, achieving sounding rocket capabilities, parabolic flights of experiments, satellite design, and satellite operation. Currently, many SA states are engaging in space activities: Bolivia, Chile, and Venezuela are operating their first satellites; Chile, Paraguay, and Uruguay are shaping their national space agencies; Ecuador is experimenting microgravity parabolic flights; Peru is designing and constructing small satellites and participating in Mars simulations; and the regional leaders, Argentina and Brazil, are developing indigenous satellites and launching vehicles as well as testing and integrating facilities. The next subsections provide a little more detail on current space activities SA states are engaging in [4].

3.1 Argentina

Argentina is currently going through a renovated golden era regarding space activity. In this country, interest in space technology started some years into the second half of the twentieth century, with a successful sound rocket program that ended abruptly in the last decade of the past century due to lack of support and international pressure. Capsules carrying living animals had also been tested with a high success rate. Since the creation of Argentina's Space Activities Committee (CONAE for its acronym in Spanish) in the '90s, space exploration turned to the civilian regime. Since then successful satellite series have been developed by CONAE, called the Satellite of Scientific Applications

(SAC for its acronym in Spanish) or SAC series. Amateur radio, universities and civil associations have conducted space experiments and/or orbited small satellites.

Today, Argentina is engaged in the development of a small satellite launcher and continues to develop earth observation satellites by its own and in cooperation with countries such as Italy (SAOCOM) and Brazil (SAC-B/SAC-C, SAC-D and SABIAMAR); and a recently created testing facility places the country as the second holder in the southern hemisphere of such facility. The recent launch of ARSAT-1, the first telecommunication satellite fully designed, built and tested within the country witnesses the current involvement of Argentina in space technology [5].

3.2 Bolivia (*Plurinational State of Bolivia*)

Bolivia passed a decree early in 2010 that created Bolivia's Space Agency (ABE for its acronym in Spanish) in the form of a state-owned enterprise under the Ministry of Public Work, Services and Housing. The agency was given the task of fulfilling the launch of Bolivia's first telecommunications satellite, as well as administering the satellite once in orbit. This satellite named Tupak Katari was built by a subsidiary of China Aerospace Science and Technology Corporation and launched from the People's Republic of China's Xichang Satellite Launch Center in late 2013 [6].

3.3 Brazil

Brazil is the regional leader in aeronautics. Embraer is the third largest airplane manufacturer, and has long been the major player in space technology. Following a long term space policy started by the nation's Armed Forces, the *Agência Espacial Brasileira* has been in charge of its management since 1994 [7][8].

Brazil has developed a series of rockets, with aims at becoming a satellite launcher, despite some minor setbacks. The country also operates two launch facilities in the southern hemisphere, *Centro de lançamento de Alcântara*, created in 1983, and *Centro de Lançamento da Barreira do Inferno*, created in 1965 [9]. Regarding spacecraft development, a long-term cooperation with China has led to CBERS (China-Brazil Earth Resources Satellite) satellite series, which comprise five earth observation satellites that provide information of the natural resources and from the vast territories of both countries [9][10]. CBERS 04A, the sixth in the role, is now under development and its launch is expected to be by the end of 2018.

An indigenous satellite Multi-Mission Platform (PMM for its acronym in Portuguese) is under development by the National Institute for Space Research (INPE for its acronym in Portuguese) and the first user of the platform will be the Brazilian

Amazonia-1 Earth observation satellite, which will complement the CBERS constellation.

Military Satellites, such as SGDC-1 and SGDC-2, will serve to secure military communications and are planned to be launched by 2016 and 2019, respectively. Both satellites had their specifications defined by *Visiona Tecnologia Espacial* (Joint-Venture Embraer-Telebras), where Thales Alenia Space is its main contractor [9]. Several Brazilian telecommunication satellites are operated by private companies occupying the country's orbital positions.

Brazil also has educational institutions, such as *Instituto Tecnológico da Aeronáutica*, the most traditional engineering college in Brazil that educates students to work in space area. In 2011, Brazilian Ministry of Education created the *Programa Ciência sem Fronteiras* (Science Without Borders Program) that gives Brazilians graduate and postgraduate students opportunities to study in top levels foreign institutions, such as MIT, Caltech, and the like. This program also has a specific research line named *Ciência sem fronteiras Espacial* (Space Science without Borders) that gives opportunities to Brazilian students to work in space centers in Canada, United States of America and Europe. The idea is to develop capable professionals to implement the Brazilian Space Program (PEB for its acronym in Portuguese).

During the history of Brazilian Space Program two huge projects have not been concluded due to lack of Brazilian financial support, such as ISS (Brazil and USA), and Alcântara Cyclone Space (Brazil and Ukraine) [7][8].

3.4 Chile

Chile has commissioned three satellites, FASAT Alfa, FASAT Bravo and FASAT Charlie, with only the latter one currently operational and delivering ground images of Chilean territory. All built under the Chilean Air Force sphere, the first two are twins, with Bravo serving as the replacement of Alfa due to separation problems. These two were built in collaboration with Surrey Satellite Technologies Ltd, while FASAT Charlie was commissioned to EADS Astrium. Chile is in the process of establishing a Space Agency as the previous body has been dissolved [11][12].

3.5 Colombia

Colombia is currently starting its space activity. In 2006 the presidency created the Colombian Space Committee (CCE for its acronym in Spanish) aiming at outlining the country's space strategy for the upcoming years with their main focus upon:

- Strengthening of knowledge and capabilities generation in science and technology: human resources, education, communication, and

knowledge broadcast; all aiming at the betterment of technological infrastructure as well as research and development to solve national problems.

- The use and application of space science and technology focusing on technology transfer, adopting space services as well as satellite development.
- International relations and cooperation.

As of 2015, the executive direction of the CCE was in charge of the Space Affairs Office of the Colombian Air Force [13]. Among other initiatives related with the aerospace field in the country are the appearance of Sequoia Space, a private company aiming to develop small and nanosatellites [14], and the program "Medellin in Space", by Ruta-N, which aims to host startups on aerospace developments [15].

Colombia also has academic initiatives. In 2007, Libertad-1, a CubeSat built by the Space Program of the Sergio Arboleda University, was launched by a Dnper rocket and became Colombia's first satellite. Currently, the Sergio Arboleda University is working on the Libertad-2 project [16]. In 2013, the Aerospace Research and Development Group of the National University of Colombia was founded. Currently, it is working on the Prometeo I mission, aiming to create the technological and organizational base for the development of launching vehicles, and aims to launch two sounding rockets [17]. Indumil, a company run by the government, has dedicated part of its research and development (R&D) resources to develop solid propellants technology.

3.6 Ecuador

Ecuador started its participation in the aerospace field with the installation of a tracking station by NASA in 1957. Yet, by its own means, the involvement of Ecuador in Space Affairs is relatively new. In 2003, the project Ecuador to Space begins, leading the path to the development of what would be later known as the EXA, the Ecuadorian Civilian Space Agency, officially constituted on November the 1st 2007, and which works closely with the Ecuadorian Air Force, FAE. Among the works of the current EXA are the astronaut training of Ronnie Nader, which took place mainly in Russia, the development of a zero-g plane, and the development of two satellites, currently in LEO. The first one, called NEE-01 PEGASO, is a 1U CubeSat, with a technology demonstration mission, weighing 2.1kg, was launched by a Long March 2D from the Jiuquan Satellite Launch Center at 04:13 UTC of April 26th 2013. This Satellite was temporarily lost by the close encounter with a Russian launcher debris cloud. The second satellite, the NEE-02 KRYSAOR, also a 1U CubeSat, had photograph and video capabilities, and was equipped with a device named PERSEUS, which succeeded in

regaining communication with PEGASO. This second Satellite was launched on November 21st, 2013 [18].

3.7 Paraguay

Paraguay has so far little to no involvement in aerospace activities. The major step towards this is the creation of the Paraguay's Space Agency (AEP for its acronym in Spanish) on March 26th, 2014 by the Law 5151. The agency has 12 institutions, including the government, military, and academy, represented by several Paraguayan entities in each of the mentioned fields. While the agency depends in function and administration of the Presidency of Paraguay, it has no funding from the government. The main goal of the AEP, as stated in the law that created it is "to promote and manage the development of national space related activities, promoting the technological innovation required to achieve and use outer space in a peaceful way". There have not yet been official announcements on projects such as rockets or satellites coming from the AEP [19].

3.8 Peru

Peru's main space programs are supervised by the National Commission of Investigation and Aerospace Development (CONIDA for its acronym in Spanish), which on December 2006 helped launch the first space Peruvian probe, Paulet, from Punta Lobos Air Force base in Pucusana [20]. The participation of Peru in the space sector has focused more on satellites, and lately in the involvement of students in Mars simulations and astrobiology.

On June 2013, Peru launched its first Peruvian-made rocket to the stratosphere, called Paulet 1-B [21]. A year later, the Peruvian Institute of Radio Astronomy developed two CubeSats for research purposes, named PUCP-Sat 1 and Pocket-PUCP, and they were launched them from the Russian Yasny launch base. Since then, two other universities have launched other CubeSats, the UAPSAT-1 and Chasqui 1. The UAPSAT-1 weighing less than 1kg was designed and programmed by students at the Universidad Alas Peruanas. It was launched on January 9th, 2014 aboard the Antares rocket out of the NASA's Wallops Flight Facility. A UAPSAT-2 nanosatellite is currently in development. Chasqui-1 nanosatellite was developed at the Universidad Nacional de Ingenieria. It was launched on February 4th, 2014 aboard Progress M-22M out of the Baikonour Cosmodrome in Russia. The nanosatellite was put into orbit out of the ISS by the crewmembers in March 2014 and since then students have been collecting data [22].

On February, 2016, CONIDA launched the National Centre of Satellite-Images Operations (CNOIS for its acronym in Spanish) with the help of the Airbus Defence and Space division. This Centre aims to

achieve technological independence regarding satellite and national security issues. The latest Peruvian satellite, PeruSAT-1, built by Airbus is scheduled to launch in September 2016 [23].

The recent Peruvian Space Week in August 2016 shows the increasing interest of Peru in getting involved in space activities. In the past few years students and professionals formed the Mars Society Team Peru to participate in the Mars Desert Research Station (MDRS) simulations in Utah, USA. They also have a dedicated group looking at astrobiology, *Sociedad de Astrobiology del Peru*, which along with the second SA-SGW coordinated the First Latin American Astrobiology Congress and International GeoBiology Forum. There is work in progress to make one of their deserts, Pampas de la Joya in Arequipa at the south of Peru, which is geologically a Mars analog [24][25].

3.9 Venezuela's

Bolivarian Agency for Space Activities (ABAE for its acronym in Spanish) is an organization ascribed to their Ministry for Science and Technology created in 2008. Its purpose is to manage space policies and peaceful uses of outer space. Current projects include:

- Telecommunication satellite VENESAT-1 (Simón Bolívar), Venezuela's first artificial satellite launched in 2008 from Xichang Satellite Launch Center by a Long March CZ-3B/E rocket.
- Earth observation satellite VRSS-1 (Miranda) launched in 2012 from Jiuquan Satellite Launch Center by a Long March 2D rocket. Its main objective is to image the national territory and improve planning of agricultural activity and response to disasters [26].

3.10 Uruguay

Uruguay is in the process of forming its space agency. Thus far an indigenous CubeSat has been developed by the *Universidad de la República's Instituto de Ingeniería Eléctrica* in collaboration with state-owned ANTEL. It launched in 2012 and represents six years of work. In addition, Uruguay holds two orbital positions, and one of them is leased to Venezuela, and Uruguay gets 10% of VENESAT-1's bandwidth in return [27].

As shown in the different SA states, their space involvement is mainly with satellites. Thus, we provide a dedicated section for it.

3.11 Satellites in South America States

South America has been known by inequality for a long time [28] and the space sector is not the exception. While some countries have a long history of efforts with varying degrees of success and impact, others have rather recently demonstrated interest and accomplished milestones in very short time; whereas others have little

or no interest or capacity to engage in space activities on their own. This is summarized in Table 1, showing each SA country's leading institution, satellites (including small satellites) and other related assets.

Table 1. Satellites and leading institutions in South American countries

		Dedicated Institution	Satellites	Ground Segment	Launcher/Sounding Rocket	Launch Sites/Integration Facilities
Argentina	Historic	CNIE(1960~1961)	7		20	
	Current (Planned)	CONAE(1991)	2(8)	6(1)	0(1)	1
Bolivia	Historic					
	Current (Planned)	ABE(2010)	1(1)	2	0	0
Brazil	Historic	GOCNAE(1961)				
	Current (Planned)	AEB(1964)	16(20)	3	1(1)/6	2/1
Chile	Historic	ACE(2001)				
	Current (Planned)	Consejo de Ministros(2013)	2(1)	1	0	0
Colombia	Historic					
	Current (Planned)	CCE (2006)	1(1)	0	0	0
Ecuador	Historic	CLISREN(1982)				
	Current (Planned)	IEE(2012), EXA(2007, civil)	2	1	0	1
Guayana	Historic	-				
	Current (Planned)	-	0	0	0	0
Paraguay	Historic					
	Current (Planned)	AEP(2014)				
Peru	Historic					
	Current (Planned)	CONIDA(1974)	4		2	1/0
Suriname	Historic	-				
	Current (Planned)	-	0	0	0	0
Uruguay	Historic					
	Current	FAU	1	1	0	0

	(Planned)					
	Historic					
Venezuela	Current (Planned)	ABAE(2005)	2	2	0	0/(1)

It is evident that the development of each nation's space sector and its commitment is very uneven; for instance, Peru's national space institution, CONIDA, was founded in 1974 as compared to Bolivia or Venezuela who have founded their national agencies less than ten years ago. However, these two countries have managed to launch at least one major satellite, built ground segment stations and have their technicians trained.

Other countries, like Brazil and Argentina, started their space ambitions earlier; they have dedicated institutions since the 1960s. Even though both national space programs have had their share of drawbacks [29][30], they have a well established system enabling domestic satellite development including design, scientific payload development, integration and testing[31][32][33][34][35]. Brazil has also attempted participation on the ISS [36][37], and had an astronaut from AEB flown to space.

Taking the first steps Colombia, Ecuador, Uruguay, and Peru have small satellites in orbit developed by independent institutions and universities [38][39][40]. With special mention of Ecuador's civilian space agency EXA, which is also engaged in microgravity parabolic flights in partnership with the Ecuadorian Air Force since 2008.

Chile has had two Earth observation satellites orbited [41][42] plus one that failed to separate from the main payload [43], all commissioned by the Chilean Air Force, and operates a ground control station but has no institution devoted to carrying national space related projects.

We propose a classification of four stages:

Stage 1: Countries that operate satellites, with satellite-building capabilities.

- Argentina
- Brazil

Stage 2: Countries that operate satellites, but have, as of May 2016 no building capabilities.

- Bolivia
- Chile
- Venezuela

Stage 3: Countries that operate and/or are in the process of building small satellites.

- Colombia
- Ecuador
- Uruguay
- Peru

Stage 4: Countries that do not operate satellites, have an incipient or no program at all.

- Guyana
- Paraguay
- Suriname

4. Regional Cooperation

4.1 Background of Cooperation in the Region

The region houses two trade blocs, Mercosur (est. 1991) and the Andean Community (est. 1969), which together with Chile, Guyana and Suriname have integrated an intergovernmental union, UNASUR in 2008. This sets a record of regional cooperation in more general areas, including technical ones such as aeronautics; establishing a suitable environment for a Regional Space Agency.

Regarding space activities early cooperation in the region can be exemplified by the *Experiencia Castor-Perú* in 1979 [44]. In 1996 Argentina and Brazil created a framework for cooperation; this was called the *Acuerdo Marco sobre Cooperación en Aplicaciones Pacíficas de Ciencia y Tecnología Espaciales* [45].

Recent examples of regional cooperation are meetings involving Bolivia and Argentina regarding cooperation in Earth observation and training of Bolivian technicians in Argentina [46], Peru successfully launched the sounding rocket Terra LM, which resulted of international cooperation between CONIDA from Peru and CONAE from Argentina [20] or the launch of an Argentinean payload on board a Brazilian made rocket [47], the planned SABIA-MAR, a binational cooperation for environmental studies of the countries' maritime coasts [48] [49] and ties established by Venezuela with Argentina, Bolivia, Brazil and Uruguay regarding training of technicians, orbital positions, and ground segment [27]. Very recently Paraguay is engaging in meetings with Argentinean contractors to develop a satellite [50]. These examples show that the players in the region have the need and the opportunity to collaborate on individual projects.

In addition to UNASUR, there are other entities that can help with the initiation and development of SASA as they are working in regional collaborations related to science and technology in general. The Andean Community of Nations manages satellite services for the Andean Community to support the process and

integration of services in telecommunication for both the prevention of natural disasters and share information from stolen mobile units in the region [51][52].

The Andean Parliament has a Commission for Education, Culture, Science, Technology and Communication targeting mainly the education advancement in different SA countries. The ones participating in this Parliament are Ecuador, Peru, Colombia, Chile, and Bolivia [53].

Another organization that is working with programs centered on science and technology is the Andres Bello Agreement. This includes Central American countries, and participating members are Bolivia, Chile, Colombia, Cuba, Ecuador, Spain, Mexico, Panama, Paraguay, Peru, Dominican Republic, and Venezuela [54].

4.2 Consensus Regarding Integration

In 2011 the then Argentine Minister of Defense, Arturo Puricelli, proposed an initiative to create a South American Space Agency [55]. This initiative was to be the responsibility of the South American Defense Council and incorporated in its *Plan de Acción 2012* but there is no record of it [56]. Even though it is unclear whether there is a document that reliably states such intentions, specialized press has informed of intentions to jointly develop an indigenous launcher [57]. In any case UNASUR would provide an existing legal framework on top of which to build such an agency, as it holds the precedent for international cooperation in the region on a large scale. It could also facilitate the implementation of common space policies, whilst allowing each member to retain some degree of independence through each national agency [58]. Also on this subject, during the seminar *Desafíos del sector espacial latinoamericano* held in Bariloche in 2014, a consensus on this matter was achieved by the participants, stating that regional cooperation is not only a possibility but a necessity [59].

South America has faced political and economical fluctuations that have prevented and delayed large scale projects throughout its history. To prevent this endeavor to fail, several aspects such as political, financial and technological should not be taken lightly. Some of these aspects are addressed in the following sections.

5. Benefits of Establishing a South American Space Agency

Space activities in South America are still modest compared to other regions. In this aspect a South American Space Agency could serve as an umbrella for the region's space programs against political and financial instability. Once a project or task has been committed from a government or governments, the project passes to be administered by the agency. Therefore, a project or mission can be fostered and

managed with more independence from political agendas and less overhead.

An added bonus to the collaborations worked through SASA is that they will provide a higher ground to its members in international negotiations. These tend to be highly technical and security sensitive, which can be expanded to other areas and used as a basis for national or inter-regional negotiations.

Countries can join forces and gather resources to promote space technology and research in academia and industry in the region. Space projects are complex and involve several areas. As one of its mandates, the agency will foster and incentivize research and development of several areas of the regional industry. For these activities to occur a legal framework to the space activity will be developed in the region.

Having an increase in research and technology development, SASA will directly and indirectly establish new job opportunities enabling high profile professionals to develop their careers locally. These professionals, with knowledge of hands on projects will, in turn, teach and train the next generation in the entire region.

As a core responsibility, SASA will coordinate larger, multinational projects that benefit the participating countries. Through this mechanism, it will facilitate cooperation among countries and research institutions such as use of laboratories and facilities, students and professional exchange programs, among others. Having as an objective to make the development of space technology more independent from the government, SASA will operate not only by facilitating the sharing of special facilities already available in the SA participating countries, but also building new facilities in different country locations necessary for the regional space program.

SASA will reinforce and create new regional alliances, and by establishing a framework simplify and incentivize cooperation and technology transfer among the region and international partners. It will help the region to develop regional space capability and in the process create new jobs and a demand for research and development in several areas to support the space program. Finally, it will foster a new generation of professionals with regional resources and training that will help the region as a whole socially, economically, and technically.

6. Challenges for the Development of a South American Space Agency and Recommendations to Address those Challenges

6.1 Legal Framework

One of the main challenges in creating SASA, involving countries with such a high discrepancy in terms of culture, economic growth, and technical

development, is to establish a cooperation frame where all the members have access to their desired processes, products, and/or services developed by the agency. For doing so, it is necessary, before anything, an agreement among all countries involved, under the form of a "technological safeguard treaty" [60], to be submitted and approved by each parliament of the participating countries. This will provide a common objective that does not interfere with national developments, focused more on the framework rather than a specific technology or development. This treaty should provide, among others, the following points:

- Confidentiality of subjects related to SASA, avoiding that know-how and technologies developed in SASA be transferred to third-parties.
- Rights on processes and technologies already developed by party-member countries at the moment of SASA establishment. In jointly developed projects, the processes and technologies developed individually by each country do not need to be shared with other entities outside SASA.
- Common access by all party-member countries to processes and technologies developed under SASA treaty.

The treaty should also provide guidelines regarding external work, equipment and services to be contracted by SASA projects. These should be assigned preferably from party-member countries whenever is possible before outsourcing it.

The creation of this new space entity would require it to be a signatory of the main legal guidelines governing space activities. Therefore, it is recommended that SASA abides by the space treaties shown in Table 2, where the Outer Space Treaty was the largest one signed in 1967. The Outer Space Treaty is formed by five treaties [61]; it specifies a common instrument or basic mechanism to resolve disputes regarding space activities, only between states. The treaty establishes the method; this corresponds to 12 initial months of negotiation. Followed by the establishment of a commission, which if it was not to find a solution would be followed by a decision of the President of the United Nations. This is what is known as conciliation mechanism. The report given by the previously mentioned commission would not be mandatory; enforcement is up to the disputing states. While the treaty contains regulations that the development of SASA would need to abide to, the challenge remains the same as for other developing countries and emerging companies. Different aspects have changed since the treaty was written in the 1960's and some stipulations require of better definition to facilitate commercialization of space and international partnerships, for example 'distribution of equitable return', 'moon utilization is not discriminatory', etc. [62]. It is worth mentioning the Moon Treaty [63],

which gives the moon and other celestial bodies the category of common heritage of mankind, thereby prohibiting exploitation by a corporation or state. This treaty was only ratified by 13 countries though, so one could assume that it has not been given much importance. Table 2 presents a short summary of the main space law treaties affecting the South American region [64].

Table 2: Main space law treaties for South America

Year	Treaty
1967	Outer Space Treaty
1968	Return of Astronauts and the Return of Objects Launched into Outer Space
1972	International Liability for Damage Caused by Space Objects
1975	Registration Convention
1979	Moon Treaty
1989	First cooperation agreement between Argentina and Brazil.
1996	Framework agreement between Argentina and Brazil
2011	South America space agency initiative was carried out by Argentine Defence sector, with a more military oriented collaboration.
2004-2011	UNASUR declaration and foundation: • Many South American countries besides Argentina and Brazil are showing interest in space activities. • Existing cooperation agreements between different South American countries

With the current treaty mechanisms among SA countries already in place, import and export regimes given by national law and MTCR can be used as basis to develop the SASA collaboration frame, though the latter would be of minor importance for a civilian space agency.

6.2 Political Aspects

One of the driven aspects that justify the creation of SASA is undoubtedly the political one, which go hand in hand with the economical aspect. Due to the extremely high cost of development and manufacture of space equipment, even developed countries have historically preferred to conduct their space programs in

cooperation with other nations. If it is true for those countries, it is even more justifiable for developing countries like those from South America.

As we have mentioned in previous sections, Brazil and Argentina present a historical experience of international cooperation in their space activities. Brazil started its major international space cooperation in 1979 with the beginning of the Brazilian Complete Space Mission (MECB for its acronym in Portuguese). The main goal was to develop its own satellite and rocket, capable to put in Earth's orbit an artifact from a launch site located in the national territory. According to [65], "the challenge, at the time, was to acquire satellite and rocketry technologies from simple projects and with international support, mainly from France, which was Brazil's partner in this project". However, "due to the dual nature (civil-military) of the rocketry technology, France gave up of taking part of this cooperation, what brought several delays to the project". Besides the dual nature of the technologies involved the fact that the Brazilian space program was under Army's responsibility also contributed to the end of that cooperation with France.

Other important international cooperation regarding the Brazilian space program was with Russia, which collaborated with the design of the Satellite Launch Vehicle (VLS for its acronym in Portuguese); China, with which Brazil conducts a profitable cooperation since 1988, for the development and manufacture of CBERS satellite series; and Ukraine, with which Brazil opened the joint-venture Alcantara Cyclone Space (ACS), that would use the Ukrainian rocket Cyclone-4 and the Brazilian launch site in Alcantara to provide commercial launch services. Brazil also cooperated with the United States of America in the dawn of the International Space Station (ISS); but aside from the cooperation with China, none of them persisted.

Brazilian option of engaging cooperation with countries as Russia, Ukraine and China certainly brought difficulties to engage a more profitable cooperation with the USA. In 2003, the Brazilian House of Representatives denied a safeguard agreement in the space area between the two countries, and Brazil has had enormous difficulties even to import components, special materials and pieces of equipment for use in CBERS satellites. Some parts have been imported only for use in the indigenous satellite Amazonia-1, yet after signing End User Statements complying that material will be used only for that purpose.

Argentina, on the other hand, made an option to establish cooperation mainly with the United States and Europe, which permitted it to establish cooperation in important programs as SAC-D Aquarius satellite, development of more complex subsystems as Attitude and Orbit Control Subsystem (AOCS), On Board Data

Handling (OBDH), and even in the development of its launch vehicle Tronador-2.

Based on the history of international cooperation in space activities involving SA countries, the establishment of SASA will certainly have impact on the current bilateral cooperation between these countries and developed countries. Depending on the kind of know-how or parts and materials provided by developed countries to SASA projects, it will be necessary to establish safeguard agreements between those developed countries and the countries involved in SASA, which could be a challenge to establish the agency.

Other political aspect that can bring some concern is how to keep the civil character of SASA between all participating countries. Brazil, for instance, in spite of having a civil space agency, has all developing activities related to its launch vehicle done by the Air Force.

6.3 Financial Sustainability

The financial sustainability of a space agency is closely linked with the political aspects. Decisions are made according to national interests, prioritizing economical stability. Often science and technology are not first concerns or priorities for some SA governments. However, some technological advancement can help further national interests. "Space is a shared resource and the international treaties ask for an equal use of the benefits, a method to decrease these initial high costs is through international partnership" [62]. A strategic orientation that correlates with long term national interests is needed. Since these interests vary in each country, a financial plan has to be developed, where efforts and resources can be located within different projects.

A possible policy to return invested resources is Geo-Return, following the model applied at the ESA, where a member state is awarded contracts depending on its financial contribution to the agency. The main advantage of Geo-Return is allowing an equitable participation of each member state, having regard to their investments [66]. Additionally, small and medium enterprises are supported through this policy by increasing their competitiveness in space sector [67]. Even if this principle represents a fair return and has multiple benefits, some lessons learned can be applied for SASA. The most important is that decisions are often not made to favor the most technological and advantageous options, leading to expensive projects. Since space budgets in many SA countries are already scarce, high expenses are to be avoided. In order to do so an investment return policy should be applied considering not only monetary return in contracts, but also opportunities to access scholarships, training programs, and certain technologies, and use of lab facilities not available in all countries. Different

member states can offer their national expertise, be it launch services, usage of testing or integration facilities, natural resources such as copper, satellite buses, or advantageous geographical locations.

The initial funding to create the organization is also a challenge due to historical conflicts that have taken place in the region. Some countries may be reluctant to contribute, due to the fear of not receiving anything in return from the agency; others may refuse to take part in the agency because they do not see the direct benefits that a regional space agency could have on their country. Therefore, a priority when creating the agency and during its administration must be financial transparency. An approach to share the expenses to create the agency is establishing a quota proportional to the gross national product of each country.

SASA should focus on the benefits of having a space industry for each specific country, in other words, showing what each country could develop and how that could fit into a regional paradigm. SASA will not only encourage the development of the aerospace industry, and high-tech in general, it also and mainly stimulates economic growth in all areas. Whether it be Ecuadorian transistors, Argentinean satellites, made of Peruvian metals for Brazilian aircraft. The agency can also show how much human capital the region would be able to retain, thanks to having a dynamic industry that would reduce the need to leave South America with the purpose of finding a job in a scientific or highly technical area. SASA should allow participating countries to place their efforts and resources within different projects. The only mandatory contribution should be for the administration of the agency and at least one project. Ideally, all countries involved contribute to the kick-off project of SASA. However, each country shall have the option to decide which projects they want to join depending on their capabilities and national interests.

We emphasize that SASA will need to implement a Geo-Return policy to fund projects considering not only a monetary return of investments but also opportunities to access certain technology, scholarships and training programs.

6.4 Cultural Aspects

While communication across cultures can be challenging and although cultural differences might seem low in South America, working with people from different professional and organizational backgrounds can be demanding. The creation of SASA will be an opportunity to improve international cooperation among all nations in the region, strengthen the potential and cultural values each member has to offer, and embrace the similarities and differences between countries to enrich the cultural background of the initiative. Several steps must be taken in this direction, to make the

transition to cooperation easier. Among others, it is suggested that in order to decrease the barriers for communication within SASA, Spanish and Portuguese shall be official languages of the agency, with English added as a third language as it is used by the majority of the scientific community. Additionally, a set of values and norms have to be developed to ensure keeping a good working environment, taking into account the cultural background of each member state. Also, specific and clear means of communication should be determined in order to decrease misunderstandings.

Given that space exploration is embedded in different human societies, the particular cultural and anthropological significance and importance of this initiative should be studied, taking into account the historical background of the SA region as a whole, and the participation of each country individually.

7. Further Implementation Recommendations

There are key members that should join the project from start of the development of SASA, those that are the most experienced nations in space affairs and have the largest economies, such as Brazil and Argentina. Having them on board a regional space agency increases the opportunities to make it happen. This agency should have a legal status, be a subject of international law, and abide all existing treaties in the *corpus iuris spatialis*.

SASA should be closely related to UNASUR, in order to avoid geopolitical conflicts between member states and concurrently pursue regional needs. However, it should not be a mere office or secretariat. It should be an independent organization, where its objectives are well defined, and working on a well structured organization. It should have a defined decision-making process and dispute settlement mechanism to lay a common ground for participating members.

The setting of such an organization represents an achievement in itself. However, we postulate that this agency should have kick-off projects to get it started and get member nations interested (see Section 7.2). Such projects can involve protection and safety of natural resources domestically developed, meteorological and Earth observation missions enabling the region to better coordinate early response on natural or anthropogenic disaster, as well as improved economical planning capabilities, starting from education programs.

7.1 Preliminary Organization

7.1.1 Responsibility and Management of Agency

Every member state should be responsible for the creation and development of the agency, each one with its experience and achievements. Since Brazil and

Argentina hold the larger development, they may be the best options to lead the effort, and possibly the management of the agency itself. A representative point of contact or organization of each participating country should be fully engaged in the preliminary meetings and discussions to ensure all needs and concerns are addressed early on. This team should include from students, young professionals to experienced professionals in government, industry and academia. The agency's management would need to be by election. Each country should get the highest possible support from their own Governments (level of ministry or entity in charge of science and technology). A long term commitment from participating countries will be required for this project to succeed.

NASA and ESA can be used as baselines, addressing South American region needs, characteristics and priorities. In addition, the creation process can be supported by the following already existing organizations, which consider technological development and cooperation among their objectives and activities, such as:

- UNASUR
- Andean Community of Nations, Satellite Services for the Andean Community
- Andean Parliament
- Andres Bello Agreement

7.1.2 *Timeframe for Development of Agency*

We propose the following timeline for the execution of the agency:

2016-2017: As a starting point, specific activities focused on sharing of current achievements and status of each country regarding space exploration and development should be carried on. Considering that there are existing organizations in each country that can deliver this goal, we consider that this can take place in a short term period.

2017-2019: Then, each country can designate an entity, commission or person in charge of participating in a preliminary team or council, representing each country, which will discuss and agree about specific milestones for creating SASA. This activity could take place in a mid-term period. In the meantime, cooperation agreements among countries should be encouraged. This designated team should also work on the structure, rules and regulations for the agency. These years would be the core stage for a successful development of the agency.

2020: SASA could be realized by 2020. This should be considered as a goal designated by the preliminary team or council created during the prior years.

7.2 *Proposed Projects of Focus*

SASA should be created considering each nation's national space plan, to see in which points (or projects) this agency could support those plans. Maybe the development of a big heavy launcher at the initial stage is not the best example; it would not answer to the region needs (Brazil and Argentina have projects of light launchers). A task this agency could tackle is to generate the right frame for the private sector to develop. In the beginning stage, the agency possibly would not have its own laboratories, universities, factories, etc, but would serve as a catalyst in the management and promotion of the existing institutions, labs, and installations to help them develop, to connect themselves, and to form a strong competitive product or project that can also be useful for the advancement of the world's space exploration program.

Three main areas that we consider as good initial projects according to the region's needs and what some countries have already being doing some progress include remote sensing, education, and open research database, which are further described below.

7.2.1 *Remote Sensing*

An initial remote sensing project at SASA offers the benefit of obtaining information of physical, chemical and/or biological properties of the SA continent. Especially, instruments on space borne platforms have the potential of covering large surface areas. Furthermore, within a remote sensing project task and responsibilities could be distributed among member states, ensuring an involvement of all SASA countries: some could be in charge of the acquisition of images and data, others could contribute to their processing, and conversion, and others could be the end users analyzing the images and presenting findings for possible protection and preparedness techniques.

Remote sensing has a wide range of applications in many different fields. The first application of interest for SASA is the management and protection of natural resources ensuring the environmental security of the continent. Monitoring land use, mapping wetland and tracking deforestation, particularly of the Amazon rainforest, would help to decide how to best protect natural resources. A second possible remote sensing project could improve meteorological observations in the region in order to have more accurate weather forecasts. A third project could include the coordination of a system for natural disaster prevention and early response. The last proposed remote sensing project comprises the implementation of an inexpensive, accurate and fairly new method analyzing satellite imagery, which allows the measurement of the economic characteristics of populations [68]. This information could assist governments when locating resources in order to reduce poverty.

7.2.2 Education

One of the important goals for SASA is the development of educational programs at different levels on the aerospace field. The Education Project can be organized at three levels. First, outreach to the society in general; second, motivational education, aimed for elementary and high school level; and third, projects held in continuing education and at all levels of higher education. The main mechanism to achieve these forms of education should not be by imposing them, but to establish goals in terms of the students' desired education, and then to make agreements with institutions already in existence in each of the countries, with capabilities to deliver such education.

Education outreach is important to inform societies of the benefits and potential of investment on the aerospace field. Projects in this field demand large budgets that are usually taken up by the government, thus this would have to be accounted for by the taxpayers in each country. Providing education outreach serves the purpose of gaining approval and support from the general public of the country and the region. Some strategies might consider partnerships with planetariums and museums to host specific aerospace events, promoting the projects held by SASA. Spinoff technologies derived from aerospace activities, and the benefits from their development should be promoted by the agency and made publicly available in each country.

The goal for the motivational education level is to encourage the involvement of students from elementary to high schools in the aerospace field. The approach for this audience would be using extracurricular activities such as: excursions to institutions related to the aerospace field in each country; exposure to documentaries, movies and similar materials, encouraging the interest in the field; and activities that provide students' access to role models who can demonstrate the viability of an aerospace-related career.

The goal of higher education and continuing education would be to form the appropriate personnel to execute the then actual and future projects of the agency and related institutions. We need to note that there is a vast range of technical, technological, and professional abilities to be acquired in order for a regional space program to be viable. The agency should be in charge of generating superior education programs taking into account the strengths of each participating country, and the necessities of the program. Advanced academic training at superior educational institutions should be encouraged to students, teachers, and other personnel by participating on the then current projects of the agency.

7.2.3 Open Research Database

A common database can help every country to perform research and promote collaboration among

students, professionals and researchers. This will allow the academic community to have a platform to share their research work and reduce the differences between the countries, as well as level their knowledge background. Some advantages are listed below:

- Create an international access point to state-of-the-art research to help develop new technologies and improve current technology corresponding to the needs and interests of the SA region
- Promote collaborations between students, professionals, universities and research institutes of SA countries
- Reduce data processing and maintenance costs
- Prevent possible duplication of work and investigation in the same topics, avoiding the unnecessary use of time and financial resources
- Develop a platform for analysis of all relevant data
- Possibility of including private industry and research activities in order to contribute to technological studies and achievements
- Allow the fields of space research to be leveled and accelerated at different SA universities and/or academic organizations
- Promote healthy competition among academic organizations and students/professionals in aerospace issues
- Contribute to the investigation of prediction of natural disasters (e.g. earthquakes, tornados), as well as other major issues that the agenda of SA countries share (security, communications, education, etc.)

Despite the existence of national regulations related to the management of digital information, some challenges and tasks are expected before the current legislation can support the needs of database development for aerospace issues. Some expected challenges are the following:

a) *Insufficient Legal Framework*

- There are some international agreements (like the Outer Space Treaty that was fully signed by Peru and Chile for example), but these are only related to government activities, leaving private activities without regulation
- Lack of adequate laws and regulations in every country

b) *Key Management Issues*

- Determination of the organization responsible for management, housing and categorization of the data
- Budget and limitations (should be ruled by SASA)
- Database accessibility: policies and procedures, timeframes for availability of data, exceptions, disclosure and terms of agreements.

c) *Institutional Weaknesses*

- Possibility of management with a short-term sight
- Threat of corruption in public and private entities related to aerospace issues
- Financial sustainability: paid and free memberships should be taken into consideration depending on the purpose and profile of users

Given these factors we provide some recommendations for the creation of a SA open research database:

- Investigate current models to use as experience and baseline (NASA, ESA, others).
- Database may be available for free to any Latin-American scholars affiliated in the program (universities, high schools and public research institutions), as well as government researchers. In contrast, any private professionals and/or companies interested in accessing the data will pay a monthly fee.
- The data stored will mainly be integrated by scientific information, satellite images, measurements and instrument manuals; and its interface should be user-friendly, for anyone to access and get easy comprehension of current space research.
- Open-source content may be fully available to anyone upon request, whereas any confidential data should be requested to the researcher institution.
- Researchers willing to submit data should follow a given procedure and template.
- Data should be available to the public for 10 years since submission. After this period, it should be moved to an external server. If the database size surpasses local storage, results should also be moved in order of submission.
- Setting the necessary conditions to properly host, index and maintain data, according to a given budget and objectives, is of high priority.
- Management and housing of database should be carried in alternated periods among countries, based on capabilities and designated by elections regulated by SASA.
- Mandatory compliance for pacific use, with national and regional security as a condition.

8. Public Outreach

As a follow up to the education project we propose for SASA to start with (explained in Section 7.2.2), public outreach is highly encouraged. Local science-related entities should put aerospace programs in the spotlight, promoting it to local elementary and high school programs, with children-oriented hands on programs to get students involved in STEM fields. Dissemination of national level direct effects,

such as: future prediction of earthquakes, management of natural disasters (know better our own geography, our country), improvement of security and communications is also advisable. Advocating and diffusion is necessary to make the development of SASA possible. This can be done by:

- Promoting regional identity and awareness of direct benefits of space activities among general public.
- Shedding light on SASA activities in non-space fields exhibiting spin-offs and spin-ins opportunities.
- Allowing the use of SA resources.
- Reviewing third party projects, e.g. university projects, start-ups, etc., and show their applications for space use and society in general.

9. Future Work

A study could be done in the cultural aspects that will take to convince the participant South American countries to remove their pride when selecting a headquarters location. If it comes to be, perhaps such location can rotate every 5 or 10 years as to avoid centralism. Having different centers according to the natural resources and/or specialties each country presents can also help with preventing centralism.

Although not part of the recommendations as initial responsibilities or applications of SASA, it would be good to start thinking about a launching site. Being that South America is near the Ecuador line, it has different options for a launching site. Local investment from the selected location would be anticipated. It would be a similar situation for ground-segment sites and test facilities. It needs to be known that whichever country hosts these facilities the accessibility and usage would need to be open to all participating countries who wish to use them upon schedules coordination. Although high initial local investment may be required, tourist opportunities can be drawn from hosting those facilities, as people from around the world may be interested in visiting them. This can be used as return of investment.

In addition to the areas recommended for the projects of the agency, a Space Agency structure would need to be well established. Providing a path for the implementation process would ensure the success of the establishment of a South American Space Agency.

10. Conclusions

A regional space agency would be a great asset to South America. It would help to further develop space technology and exploration, new business opportunities, and science, technology, and industry while developing highly trained professionals. It should aim to coordinate regional efforts in the space sector, promote and facilitate cooperation between member countries,

promote education in science and technology, and generate job opportunities for professionals in the space field.

The case of a South American Space Agency would bring advantages to its participant members. Challenges are expected during its implementation process as well as while running such an intergovernmental entity. Yet, having a well established path for the realization of the agency with the involvement of all participant nations from the planning phase would ensure its successful operations. Perhaps the first step for all participant SA countries would be to assign a good amount of their yearly national budget to science and technology as a verification of their commitment to the agency and the benefits it will bring to all the nations.

SASA would serve as an umbrella for SA space programs, shielding them against political instability. Advantages for creating such agency include: giving SA countries a better position to negotiate with other space agencies, promote space technology and research in academia and industry in the region, provide a legal framework to the space activity in the region, and take advantage of the resources in the region. SASA would coordinate big projects that cannot be carried out by a single country such as the management and maintenance of an Open Space Research Database, and facilitate cooperation among countries such as the utilization of laboratory facilities, students exchange programs, and the like.

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References

[1] B. Sarli., J. Silva-Martinez, K. Toole, A. Demeubayeva, Chibueze, J. Space Generation: Engaging the Next Generation in the Asia Pacific and South American Regions, IAC-

15,E5,6,4,x31415, 66th International Astronautical Congress, Jerusalem, Israel, 2015, 12-16 October.
[2] Space Generation Advisory Council, South American Regional Space Generation Workshop 2015, <http://spacegeneration.org/event/past-events/sa-sgw-2015.html>, (accessed 08.06.16).
[3] Space Generation Advisory Council, South American Regional Space Generation Workshop 2016, http://spacegeneration.org/index.php?option=com_content&view=article&id=1649&catid=124&Itemid=2837, (accessed 08.06.16).
[4] B. Sarli, M. Cabero, et al., South American Space Era, IAC-15-E4.2, 66th International Astronautical Congress, Jerusalem, Israel, 2015, 12-16 October.
[5] NASA, ArSat-1: Argentina to communicate its heightened space ambitions, <http://www.nasaspacesflight.com/2014/10/arsat-1-argentina-communicate-heightened-space-ambitions/>, (accessed 08.18.16).
[6] SpaceNews, China Launches Bolivia's First Telecom Satellite, <http://spacenews.com/38800-china-launches-bolivas-first-telecom-satellite/>, (accessed 08.18.16).
[7] A política espacial brasileira (parte I). Brasil: Câmara dos Deputados - Conselho de altos estudos e avaliação tecnológica, Brasília, 2010.
[8] A política espacial brasileira (parte II - Análises Técnicas). Brasil: Câmara dos Deputados - Conselho de altos estudos e avaliação tecnológica, Brasília 2010.
[9] PNAE: Programa Nacional de Atividades Espaciais 2012-2021, Agência Espacial Brasileira. Brasil: Ministério da Ciência, Tecnologia e Inovação, Agência Espacial Brasileira, 2012.
[10] F. de Oliveira, Brasil-China-20 anos de cooperação espacial: CBERS - O satélite da parceria estratégica. São Carlos, SP. Editora Cubo, 2009.
[11] Diputados piden reponer Agencia Chilena del Espacio, <http://radio.uchile.cl/2014/07/18/diputados-piden-reponer-agencia-chilena-del-espacio/>, (accessed 08.12.16).
[12] Chile: Impulso a la creación de una Agencia Espacial, <http://latamsatelital.com/2016/05/13/chile-impulso-a-la-creacion-de-una-agencia-espacial/>, (accessed 08.12.16).
[13] Comisión Colombiana del Espacio, Alcance de la CCE, <https://www.cce.gov.co/alcance>, (accessed 08.12.16).
[14] Sequoia Space, Un programa educativo de Sequoia Space, <http://www.sequoiaspace.com/>, (accessed 09.06.16).
[15] Medellín Espacial, Un programa para científicos y emprendedores de la industria aeroespacial, <http://rutanmedellin.org/es/ciudadanos/item/medellin-espacial>, (accessed 08.14.16).

- [16] R. Joya, Libertad 1, Primer Satélite Colombiano en el Espacio. *Innovación y Ciencia*, 2007, 14:16-23.
- [17] N. Alvarez, R. Huérano, O. Ojeda, Diseño e implementación de misión para el lanzamiento de un cohete para tres kilómetros de altura, VIII Congreso Argentino de Tecnología Espacial, Buenos Aires, Argentina, 2015, 6-8 May.
- [18] EXA, Agencia Espacial Civil Ecuatoriana, <http://exa.ec/>, (accessed 09.06.16).
- [19] Ley N° 5.151 de Agencia Espacial del Paraguay, <http://www.escritosdederecho.com/2014/06/ley-5151-del-26-03-2014-agencia-espacial-del-paraguay.html>, (accessed 09.06.16).
- [20] J.M. Canales, Sounding Rocket Program in Peru, SpaceOps 2012 Conference Stockholm Sweden.
- [21] Space Travel Exploration and Tourism, Peru launches first homemade rocket, http://www.space-travel.com/reports/Peru_launches_first_homemade_rocket_999.html, (accessed 09.19.16)
- [22] SGAC Peru, <http://www.spacegeneration.org/sgac-regions/south-america/peru.html>, (accessed 09.01.16)
- [23] Satnews Daily, PerúSAT-1 Passes Milestone and Fit For Flight, August 2016.
- [24] J. Valdivia, R. Navarro, F. Ortega, L. Fletcher, S. Perez, R. Condori, C. McKay, Multidisciplinary approach of the hyperarid desert of Pampas de La Joya in southern Peru as a new Mars-like soil analog, *Geochimica et Cosmochimica Acta* 75, pp. 1975–1991, 2011.
- [25] M. Abarca, A. Diaz, H. de las Casas, A. Lazarte, S. Trujillo, E. Onchi, Establishment of a Mars Analogue Research Station in Pampas de la Joya (Atacama Desert in Southern Peru), *The Mars Society*, 2013.
- [26] Agencia Bolivariana para Actividades Espaciales, Descripción e Historia, <http://www.abae.gob.ve/web/VRSS-1.php>, (accessed 09.06.16).
- [27] Agencia Bolivariana para Actividades Espaciales. Cooperación Internacional de la Agencia Bolivariana para Actividades Espaciales <http://www.abae.gob.ve/web/ConveniosInt.php>, (accessed 09.06.16).
- [28] G. Eduardo, Las venas abiertas de América Latina, Monthly Review, ISBN:950-895-094-3, 1971.
- [29] Relatório da Investigacao do Acidente Ocorrido Com O VLS-1 V03, EM; Alcantara Maranhao. Servicio Publico Federal Ministerio de Defesa, Comando da Aeronautica Departamento de Pesquisas e Desenvolvimento, 22 August 2003, http://www.aereo.jor.br/downloads/VLS-1_V03_Relatorio_Final.pdf, (accessed 08.26.16).
- [30] P. De León, Historia de la Actividad Espacial en la Argentina II: El proyecto Cóndor, ISBN: 978-987-33-7359-6, 2015.
- [31] INPE, Laboratorio de Integracao e Testes. About <http://www.lit.inpe.br/sobre>, (accessed 08.26.16).
- [32] INVAP, 40 Anos, Area Espacial y Gobierno, <http://www.invap.com.ar/es/espacial-y-gobierno/area-aeroespacial-y-gobierno/introduccion/609-presentacion-del-area-aeroespacial-y-gobierno.html>, (accessed 08.26.16).
- [33] Centro de Ensayos de Alta Tecnología, Sobre CEATSA, <http://www.ceatsa.com.ar/empresa/sobre-ceatsa>, (accessed 08.26.16).
- [34] CONAE, Centro Espacial Teófilo Tabanera, <http://www.conae.gov.ar/index.php/espanol/centro-espacial>, (accessed 08.26.16).
- [35] L. Drewes, El sector espacial argentino: Instituciones referentes, proveedores y desafíos, ARSAT-Empresa Argentina de Soluciones Satelitales, ISBN 978-987-45569-0-5, 2014, <http://www.arsat.com.ar/web/wp-content/uploads/El-sector-espacial-argentino.pdf>, (accessed 08.26.16)
- [36] NASA Signs International Space Station Agreement with Brazil, http://www.nasa.gov/centers/johnson/news/releases/1996_1998/h97-233.html, (accessed 08.26.16).
- [37] Made in Brazil O Brasil na Estação Espacial Internacional, <http://gizmodo.uol.com.br/made-brazil-o-brasil-na-estacao-espacial-internacional/>, (accessed 08.26.16).
- [38] Agencia Espacial Civil Ecuatoriana, El 25 de Enero de 2014 la señal del NEE-01 Pegaso fue recuperada por el NEE-02 Krysaor, <http://pegaso.exa.ec/>, (accessed 08.26.16).
- [39] EXA, NEE-02 KRYSAOR 2° Satélite Ecuatoriano, <http://exa.ec/krysaor/index.htm>, (accessed 08.26.16).
- [40] Antel, Antelsat, <http://www.antel.com.uy/antel/personas-y-hogares/antelsat/antelsat-ampliacion>, (accessed 08.26.16).
- [41] FASat-Bravo: The Mission, <https://www.sstl.co.uk/Missions/FASat-Bravo-Launched-1998/FASat-Bravo/FASat-Bravo-The-Mission>, (accessed 08.26.16).
- [42] Observación de la Tierra de alta resolución para Chile, <http://www.space-airbusds.com/es/programas/ssot.html>, (accessed 08.26.16).
- [43] FASat-A: The Mission, <https://www.sstl.co.uk/Missions/FASat-A-Launched-1995/FASat-A/FASat-A-The-Mission>, (accessed 08.26.16).
- [44] P. De León, Historia de la Actividad Espacial en la Argentina. Tomo I, ISBN: 978-0-557-01782-9, 2008.
- [45] Ministerio de Justicia y Derechos Humanos, Presidencia de la Nación-Argentina, Informacion Legislativa, Ley 24.925: Apruébase un Acuerdo Marco de Cooperación en Aplicaciones Pacíficas de Ciencia y Tecnología Espaciales suscrito con el Gobierno de la República Federativa del Brasil, <http://servicios.infoleg.gob.ar/infolegInternet/anexos/45000-49999/48480/norma.htm>, (accessed 08.30.16).

- [46] Ministerio de Comunicación, Bolivia y Argentina se reúnen para impulsar cooperación espacial en satélites de prospección <http://www.comunicacion.gob.bo/?q=20140902/16565>, (accessed 08.26.16).
- [47] Comisión Nacional de Actividades Espaciales, Logro de la Cooperación Espacial entre Argentina y Brasil, Exitoso Lanzamiento Suborbital <http://www.conae.gov.ar/prensa/brasil.html>, (accessed 08.26.16).
- [48] Ministério da Ciência e Tecnologia, Instituto Nacional de Pesquisas Espaciais, Plano Diretor 2011-2015, São José dos Campos, Julho de 2011, http://www.inpe.br/noticias/arquivos/pdf/Plano_diretor_miolo.pdf, (accessed 08.26.16).
- [49] Comisión Nacional de Actividades Espaciales, <http://www.conae.gov.ar/index.php/espanol/misiones-satelitales/sabiamar/objetivos>, (accessed 08.26.16).
- [50] En Parque Tecnológico de Itaipu se incubía primer satélite de Paraguay, <http://www.ip.gov.py/ip/?p=113166>, (accessed 08.26.16).
- [51] Comunidad Andina, Telecomunicaciones, [#](http://www.comunidadandina.org/Seccion.aspx?id=72&tipo=TE&title=telecomunicaciones), (accessed 09.04.16).
- [52] Comunidad Andina, Observatorio Andino de Tecnologías de Información y Comunicación, [#](http://www.comunidadandina.org/obatic/Default.aspx?idmenu=230&idpais=0), (accessed 09.04.16).
- [53] Parlamento Andino, Parlamento Andino Oficina Nacional, <http://www.parlamentoandino.org/index.php/acerca-de/organos/comisiones/comision-segunda>, (accessed 09.02.16).
- [54] Convenio Andres Bello, Secretaria Ejecutiva del Convenio Andres Bello, Ciencia y Tecnología, <http://convenioandresbello.org/inicio/ciencia-y-tecnologia/>, (accessed 09.02.16).
- [55] MercoPress, Argentina with Brazilian support proposes a South American Space Agency, <http://en.mercopress.com/2011/09/01/argentina-with-brazilian-support-proposes-a-south-american-space-agency>, (accessed 09.06.16).
- [56] Ministerio de Defensa del Perú, UNASUR Ejes del Plan de Acción del CDS, <http://www.perucdsunasur.org.pe/index.php/servicios>, (accessed 08.26.16).
- [57] Los ministros de Defensa de Unasur plantean crear una agencia espacial conjunta, <http://www.infoespacial.com/latam/2011/11/13/noticia-los-ministros-de-defensa-de-unasur-plantan-crear-una-agencia-espacial-conjunta.html>, (accessed 08.26.16).
- [58] Unión de Naciones Suramericanas, Objetivos Específicos. <http://www.unasursg.org/es/objetivos-especificos>, (accessed 08.14.16).
- [59] Instituciones del Sector Espacial de Latinoamérica analizaron en Bariloche Posibles Vías de Colaboración Regional http://www.arsat.com.ar/wp-content/uploads/2014/12/INSTITUCIONES-DEL-SECTOR-ESPACIAL-DE_LATINOAM%C3%89RICA.pdf, (accessed 08.26.16).
- [60] M.C. Mineiro, Space Technology Export Controls and International Cooperation in Outer Space, Space Regulations Library, Springer Netherlands, pp. 3-29, 2012. DOI: 10.1007/978-94-007-2567-6.
- [61] United Nations Office for Outer Space Affairs, United Nations Treaties and Principles on Outer Space Affairs, General Assembly 21st session, RES 2222 (XXI), 1966, http://www.unoosa.org/pdf/gares/ARES_21_2222E.pdf, (accessed 09.06.16).
- [62] J. Silva-Martinez, Exploring the Viability of Space Solar Power, AIAA Space 2014 Conference & Exposition, DOI: 10.2514/6.2014-4335. San Diego, CA, 2014, 4-7 August.
- [63] United Nations Office for Outer Space Affairs, Agreement Governing the Activities of States on the Moon and Other Celestial Bodies, General Assembly 34th session, RES 34/68, 1979, http://www.unoosa.org/pdf/gares/ARES_34_68E.pdf (accessed 09.06.16).
- [64] Committee on the Peaceful Uses of Outer Space, Status of International Agreements relating to activities in outer space as at 1 January 2016, A/AC.105/C.2/2016/CRP.3, Legal Subcommittee, Fifty-fifth session, Vienna, 4-15 April 2016, http://www.unoosa.org/documents/pdf/spacelaw/treatystatus/AC105_C2_2016_CRP03E.pdf, (accessed 09.06.16).
- [65] Desafios do programa espacial brasileiro. Brasil: Presidência da República - Secretaria de Assuntos Estratégicos, Brasília, 2011.
- [66] ESA, Industrial Policy and Geographical distribution, August 2006, http://www.esa.int/About_Us/Business_with_ESA/How_to_do/, (accessed 08.30.16).
- [67] J. Lahtinen, SME view on project marketing - case study in the context of institutional space business, Master's thesis, 2013 http://epub.lib.aalto.fi/en/ethesis/pdf/13427/hse_ethesis_13427.pdf, (accessed 09.06.16).
- [68] N. Jean, M. Burke, M. Xie, W.M. Davis, D.B. Lobell, S. Ermon, Combining satellite imagery and machine learning to predict poverty, <http://science.sciencemag.org/content/sci/353/6301/790.full.pdf>, (accessed 09.06.16).